Vaccination strategies on a proximity network

An agent-based outbreak on the Copenhagen Networks Study

Group __ · names · course code

Data: Sapiezynski et al., Scientific Data 6, 315 (2019)

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The network

Copenhagen Networks Study: ~700 first-year students, 4 weeks

Every phone scans for nearby phones every 5 minutes

692 nodes, 2,426,279 Bluetooth scans

Signal strength stands in for distance (RSSI)

Close contact = at least -75 dBm, about 1.5 m

Accessed through Netzschleuder: networks.skewed.de/net/copenhagen

Network statistics

Close-contact ties (4 weeks aggregated): 23,670

Mean degree 68.4 · median 65 · max 214

Density 0.099 · giant component 673 of 692

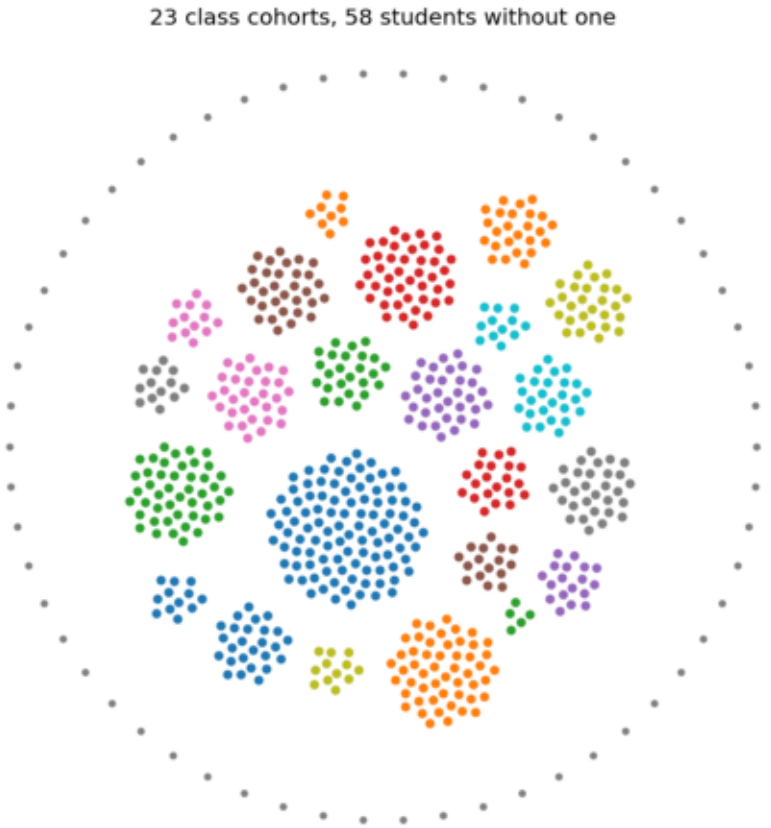
Average shortest path 2.02 · diameter 5

Clustering 0.28, against 0.18 when rewired

Degree assortativity +0.08 · deepest core: 49-core, 337 students

two thirds of ties are under 30 minutes in total

Class cohorts



23 cohorts of 4+ students, largest 104; 58 students without one

Research question

With doses for a tenth of the students, whom should we vaccinate?

Does targeting by contact time beat targeting by number of contacts?

Does the ranking survive a change in how the disease spreads?

The model

Walks the real contacts in time order, five minutes at a time

Close contact costs the other person 5 minutes of immunity

Sharing a room costs 0.8 minutes

Immunity drawn per person: 2.5 to 30 hours of exposure

Infectious for 5 days, then drops out

Each night a chance of exposure at home

No latent period, no symptoms, no variation in infectiousness between people

The eight strategies

Random · Acquaintance (a random contact of a random person)

Degree · Contact time · Betweenness · Community bridges

Week-1 degree (only past data) · Ring (reacts to detected cases)

Budget: 10% of students, vaccinated before the outbreak

except ring, which spends the same budget during it

Monte Carlo design

A starting configuration fixes patient zero, the start time,

every person's immunity and the nightly home draws

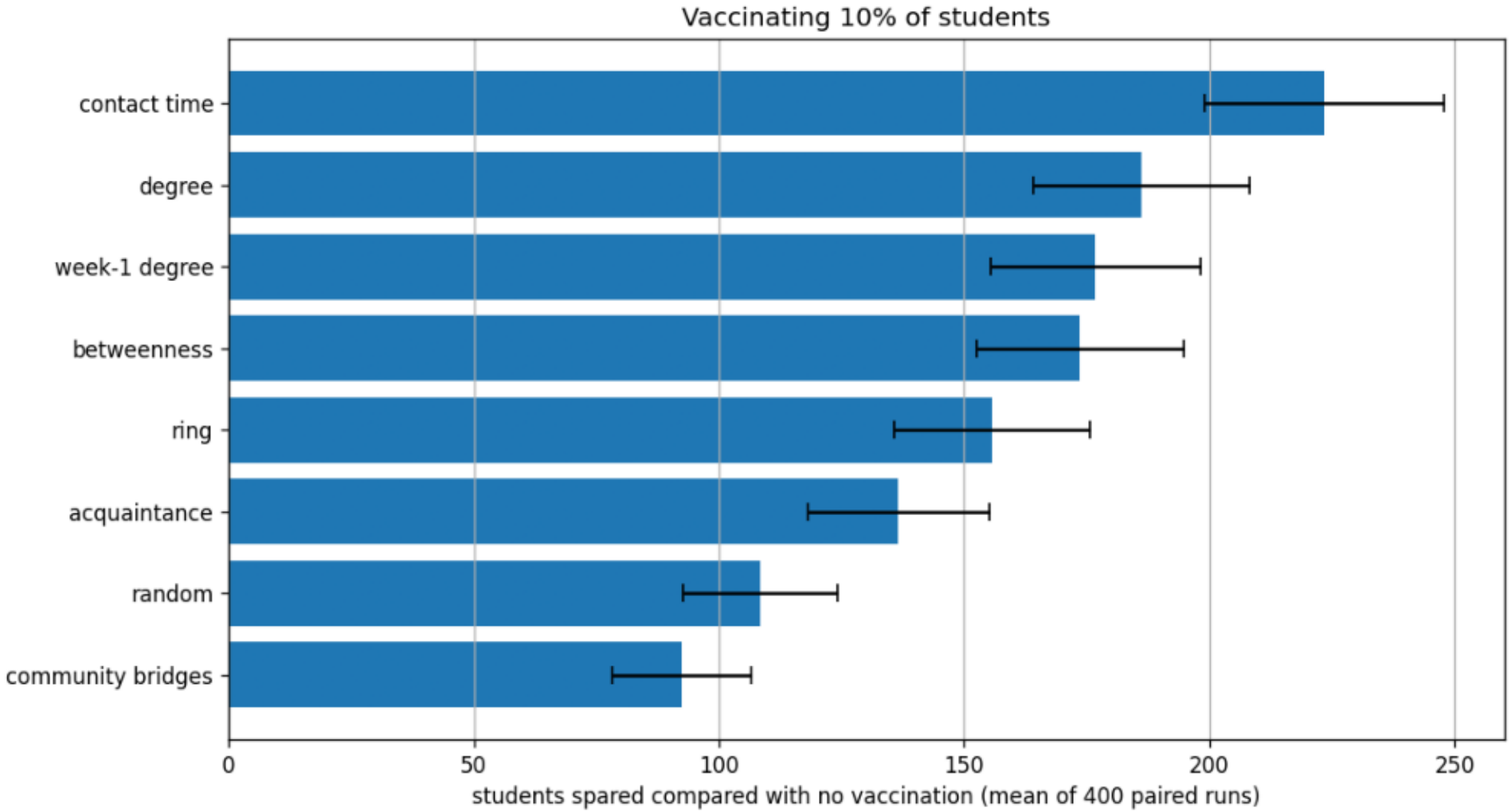
Every strategy runs on the same 400 configurations

Reported: mean infected, share that took off (50+), and

'saved' = people spared versus no vaccination, same configuration

3600 simulations behind the table on the next slide

Which strategy saves most



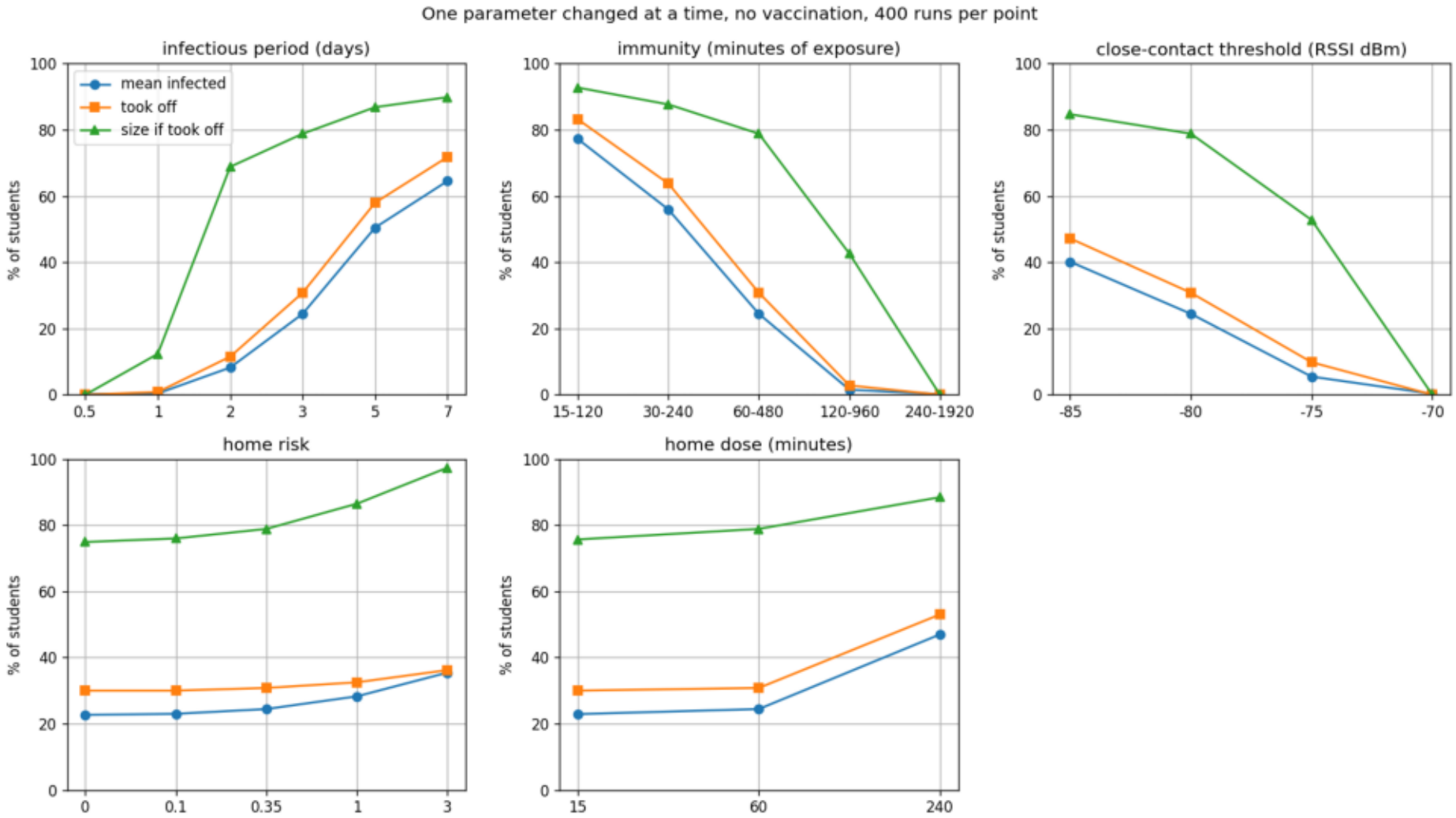
contact time saves 223 ± 24 students; random saves 108

Results table

strategy	mean infected	took off	saved
contact time	17.3%	36.5%	223.4
degree	22.7%	40.5%	186.1
week-1 degree	24.1%	42.2%	176.8
betweenness	24.5%	42.2%	173.6
no vaccination	49.6%	58.8%	

400 paired configurations, 10% coverage, current parameters

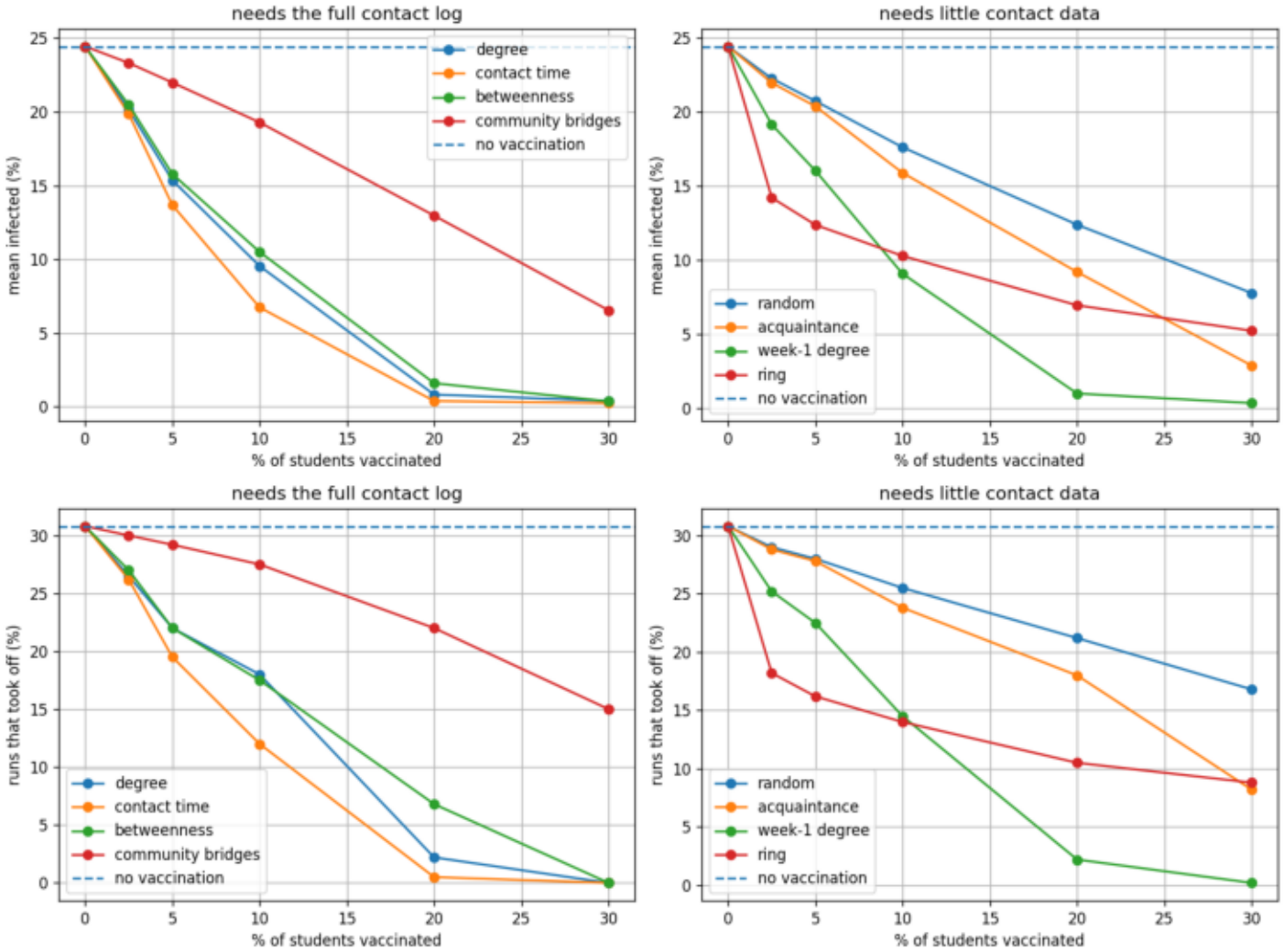
What the disease parameters do



30,000 runs: thresholds dominate, not smooth trends

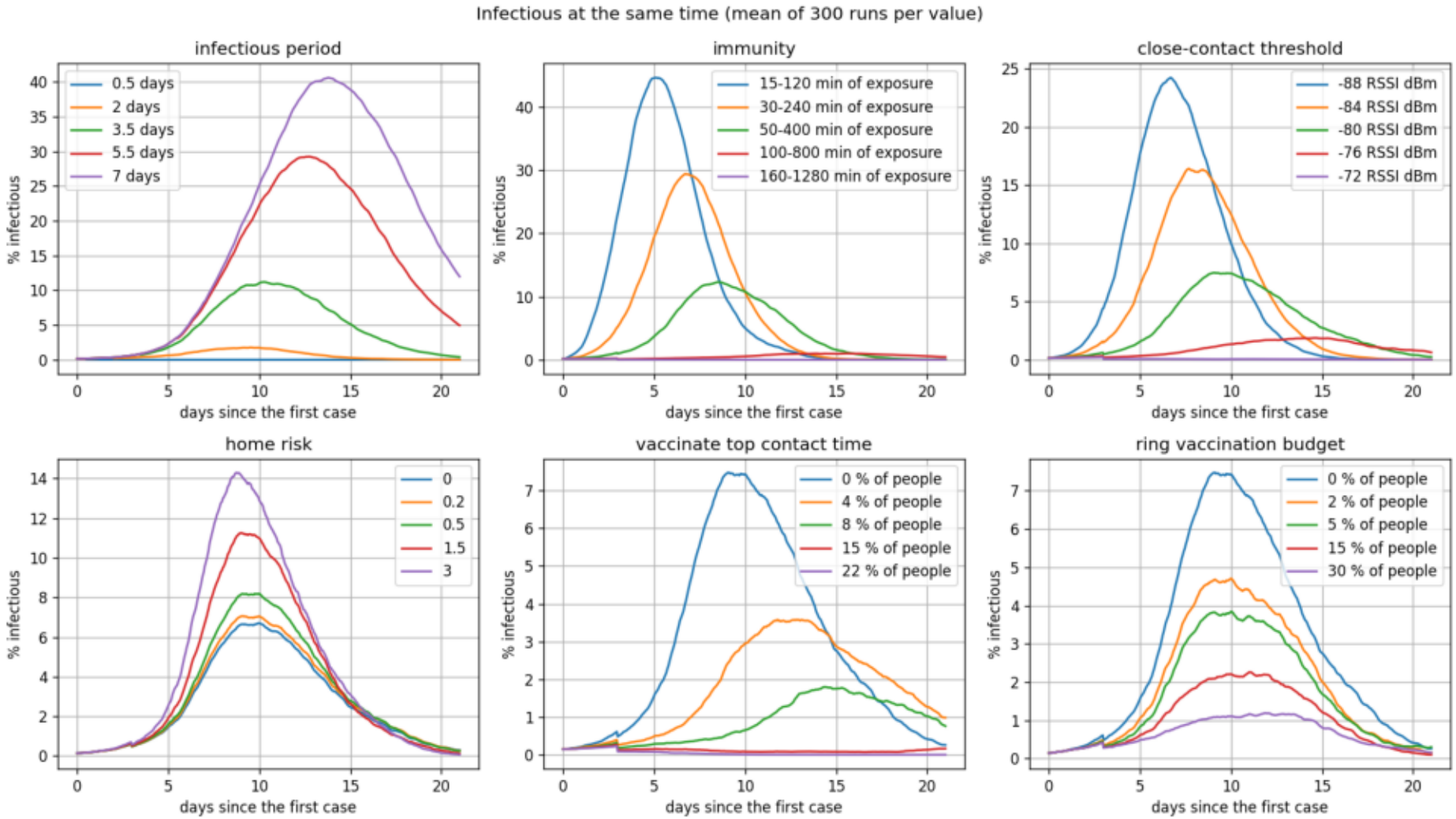
Sweeps use the simpler baseline disease, not the COVID-like parameters

Strategies against coverage



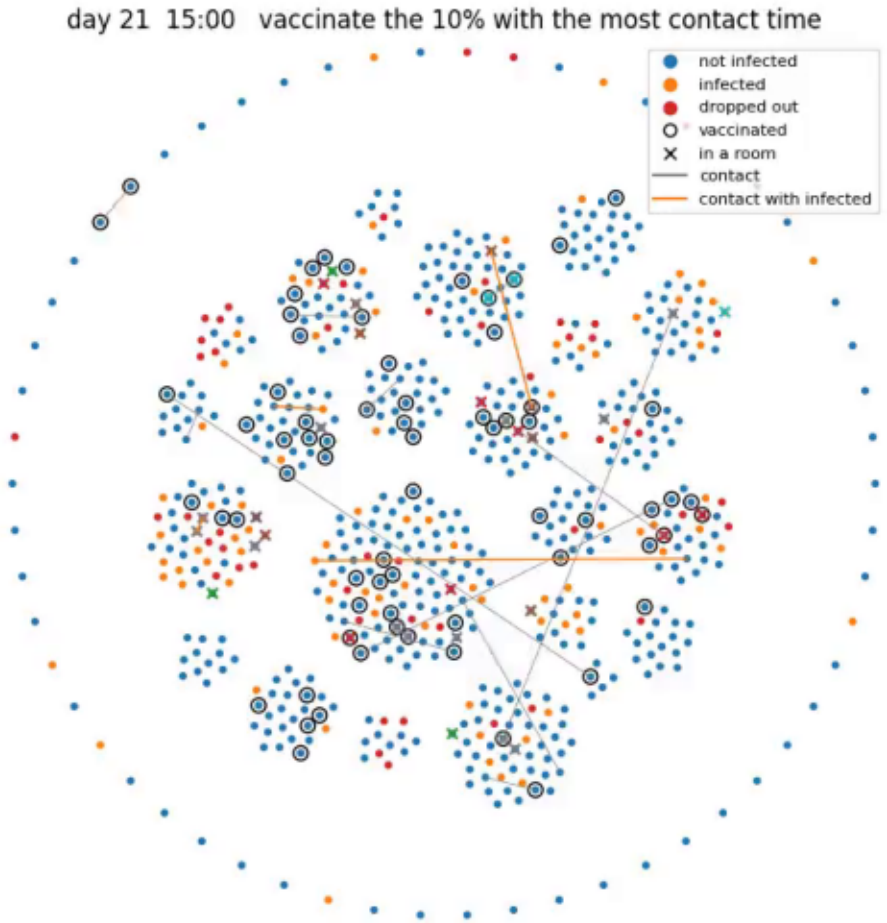
between 10% and 20% coverage, targeted strategies stop the outbreak

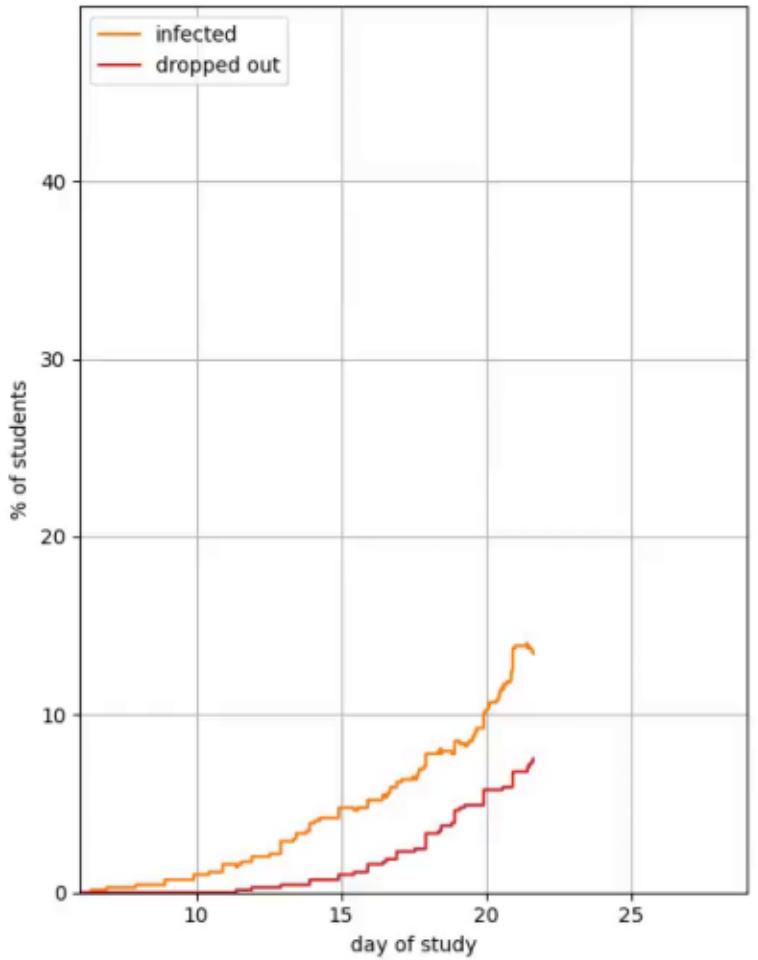
Epidemic curves



18,000 runs; the peak, not the total, is what strategies flatten

The outbreak, frame by frame





same first case, with and without vaccination

Video: covid_best_strategy.mp4 and covid_no_vaccination.mp4

Deep dive 1

TO BE DECIDED BY THE AUTHORS

Deep dive 2

TO BE DECIDED BY THE AUTHORS

Sensitivity and robustness

TO BE DECIDED BY THE AUTHORS

Limitations

Bluetooth is a noisy proxy for distance; the threshold changes everything

Contacts with people outside the study were removed from the data

Four weeks caps how far an outbreak can run

No latent period, no symptoms, no superspreading

Vaccines here are complete and immediate

Conclusion

Targeting the students with the most contact time saves the most people

One week of past contacts is nearly as good as four

Reactive (ring) vaccination depends on how fast the disease moves

Open question for the deep dives: does the ranking hold when

the disease, the contact threshold or the budget change?

References

[1] Sapiezynski, Stopczynski, Lassen, Lehmann (2019).
Interaction data from the Copenhagen Networks Study. Scientific Data 6, 315.

[2] Sekara, Lehmann (2014). The strength of friendship ties in
proximity sensor data. PLoS ONE.

[3] Netzschleuder network catalogue, networks.skewed.de

Python, NumPy, matplotlib, graph-tool